

Vincent Do

Berkeley, MO | 779-207-6818 | vmd306.com | [linkedin.com/in/vincent-do-uiuc](https://www.linkedin.com/in/vincent-do-uiuc) | vincentdo306@gmail.com

EXPERIENCE

Boeing Company

Berkeley, MO

Software Engineer

Aug 2024 – Present

- Developed production C++ features delivering real-time message streams to embedded display systems, with automated tests and Dockerized build workflows for deployment
- Owned 3+ integration and release readiness for a shared framework consumed by 60+ dependent modules, resolving compatibility issues to maintain timing accuracy in real-time message delivery
- Identified root cause of long-standing cross-system defect within two weeks of onboarding to unfamiliar systems, tracing a binary data merge edge case and restoring correct downstream output behavior
- Spearheaded month-long evaluation of Lockheed Martin software through custom integration with messaging architecture, verifying customer requirements at 1.5x the required message rate and documenting procedures for future releases

Expedia Group

Chicago, IL

Software Development Engineer Intern

May 2023 – Aug 2023

- Built customer-facing Expedia mobile search features, including VIP Access badges, sort/filter options, and improved error handling to reduce booking search friction
- Implemented GraphQL schema changes across multiple Kotlin microservices, enabling new mobile booking flows across backend and client surfaces
- Built backend support for localized search components across 16 languages, partnering with frontend and localization teams to validate dynamic rendering before release

University of Wisconsin–Madison

Madison, WI

Biomedical Undergraduate Researcher

May 2021 – Aug 2021

- Analyzed 16,813 genes across 169 phenotypes using Python/R, graph traversal, and visualization to identify phenotype-gene associations
- Reduced search space to 25 high-priority Zellweger candidate genes using frequency thresholds, chromosomal-location filtering, and recursive phenotype hierarchy traversal

PROJECTS

Playable Chess AI | *Python, PyTorch, CUDA, Flask, Pygame, Stockfish*

[playable-chess-AI](#)

- Trained CNN+LSTM legal move-policy model on 4.18M GM/Magnus positions, achieving 71.2% top-5 accuracy on a 390K-position held-out test set
- Engineered GPU-accelerated PyTorch training pipeline with mixed precision and a custom IterableDataset, reducing per-epoch training time from 20+ hours on CPU to 15–25 minutes
- Engineered PGN preprocessing pipeline for zipped datasets, deduplicating games, computing centipawn-loss metadata, generating legal-move masks, and producing compressed train/validation/test chunks

Shortest Path Between Airports | *C++, React*

[airports-global](#)

- Implemented BFS, Floyd–Warshall, and betweenness centrality algorithms to analyze connectivity and routing patterns in aviation networks, visualizing top 10 most important airports using WebGL-based globe rendering

Shafa API | *TypeScript, Hono.js, Drizzle ORM, PostgreSQL, Neon, Cloudflare Workers*

[shafa.app](#)

- Built paginated REST API endpoints for items and outfits using Hono.js, Drizzle ORM, PostgreSQL/Neon, and Cloudflare Workers
- Added backend smoke tests and local database test tooling with Jest/Wrangler to improve API validation

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science & Chemistry — GPA: 3.6

Aug 2020 – May 2024

TECHNICAL SKILLS

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL, Kotlin

Frameworks/Tools: React, Node.js, Flask, PyTorch, CUDA, REST APIs, GraphQL, Docker, Jenkins, Git, Linux, MongoDB, PostgreSQL, GCP